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Synthesis and tribological studies of epoxidized palm stearin methyl ester as a green lubricant

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ABSTRACT

Most of the base oils used in lubricant formulations are typically derived from mineral oils, which are inherently harmful to the environment. For this reason, the aim of this study was to find an alternative base oil that can be synthesised from a renewable and biodegradable source and, most importantly, that is not a threat to the ecological environment. In this paper, palm stearin methyl ester, a local oil source, was utilised as the initial raw material in an epoxidation reaction. The oil was structurally modified at its carbon double bonds using acetic acid and hydrogen peroxide (H_2O_2) in the presence of lipase as biocatalyst. Response surface methodology was used to investigate and optimize the effects of epoxidation variables, such as lipase concentration, temperature, and H_2O_2/acetic acid/C=C molar ratios on the relative oxirane conversion (ROC) of PSME. An optimum ROC of 98.9 ± 0.6% was obtained for the epoxidized PSME at a H_2O_2/C=C molar ratio of 1.51 mol, reaction time of 4.34 h, temperature of 52 °C, enzyme concentration of 6.416 wt%, and acetic acid/C=C molar ratio of 0.163:1. From the tribological studies, at each sliding speed and temperature condition, the produced EPSME bio lubricants consistently demonstrated lower friction coefficients and wear scar diameter values than those of PSME. The maximum reductions in COF and WSD values were found at 1200 rpm (44.5%) and 95 °C (16.6%), respectively.

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1. Introduction

Since the beginning of the twentieth century, the ultimate goal in every corner of the globe has been the development of green products with the main intention of conserving the earth's resources and minimizing the effects of global warming as much as possible. Lubricant manufacturing is one of the industries that consume a huge amount of fossil resources since it has broad applications in equipment, machinery and transportation. As such, determined efforts have been made with regard to the production of bio-additive and bio-based oils for the formulation of lubricants (Ho et al., 2019; Karmakar et al., 2017; Kleinaitė et al., 2014; McNutt and He, 2016; Salimon et al., 2014; Syahir et al., 2017). Biolubricants are feasible to be employed in various industrial applications such metalforming, machining, offshore drilling, two-stroke engines and agriculture machinery (Panchal et al., 2017; Reeves et al., 2017). The use of biolubricants will not only improve the efficiency of systems but may also result in the reduction of energy consumption and CO_2 emissions globally (Holmberg and Erdemir, 2017; Wong and Tung, 2016). Epoxidized vegetable oil, which is considered to be a biodegradable, environmentally friendly, renewable and less-polluting product, has demonstrated a high potential to be a substitute for the base oil in the formulation of lubricants compared to its natural state because of its high tolerance towards oxidation and thermal degradation (Panchal et al., 2017). Traditionally, the use of a mineral acid like sulphuric acid or hydrochloric acid as a catalyst in the epoxidation reaction of vegetable oil would lead to the corrosion of the walls of the equipment, difficulty in the separation of the water-soluble mineral acid, and undesirable side reactions, such as the opening of rings (Guldhe et al., 2015; Meher et al., 2006; Milchert et al., 2015). Therefore, the enzymatic route, which is regarded as a clean and green method, is perceived to be a promising alternative to the chemical route because of the mild reaction temperature, high conversion rate and selectivity, and less formation of by-products (Narwal and Gupta, 2013). Furthermore,

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implementation of enzymatic reaction at industry led to a reduction in the emission of greenhouse gas like CO_2, NO_x and SO_x, formation of photochemical ozone, eutrophication and consumption of energy (Jegannathan and Nielsen, 2013).

In the chemoenzymatic epoxidation reaction, the catalytic function of the lipase involves two actions: firstly, the lipase acts as a per-hydrolase to produce peracids, followed by the epoxidation of the carbon double bonds (Aouf et al., 2014). A conversion of approximately 83% of sapindus mukorossi oil into epoxidized oil was achieved in 8 h at 30 °C using 15 wt% of Candida sp 99−125 (Zhang et al., 2017). Novozym 435 was used for the enzymatic epoxidation of soybean oil, and a huge conversion of 96.4% was acquired in 5 h at 50 °C in the presence of the enzyme at a concentration of 3.0 wt% (Bajwa et al., 2016). (Sun et al., 2011b) obtained epoxidized corn oil with a conversion of 97.94% within 10 h at 35 °C by employing 3 wt% of Novozym 435. (Liu et al., 2016), (Chen et al., 2017) and (Bhalerao et al., 2018) also employed a similar enzymatic route for the epoxidation of sunflower oil and soybean oil, respectively. From the previous investigations, it can be inferred that a high conversion rate and epoxy content are predominated by the reaction temperature, time, lipase concentration, and molar ratios of C=C/H_2O_2 and C=C/carboxylic acid. In other studies, some researchers utilised response surface methodology (RSM) to further optimize the experimental variables in order to maximize the conversion efficiency (Zheng et al., 2018). reported that a high conversion rate of 92% of waste cooking oil methyl ester into oxirane rings was acquired under optimum conditions as follows: C=C/H_2O_2/formic acid molar ratio of 1.00/0.35/2.41, reaction temperature of 65.84 °C and reaction time of 7 h. In another RSM study, a maximum epoxy content in waste cooking oil methyl ester was obtained under optimum conditions (catalyst concentration of 28.17%, reaction time of 7.51 h, temperature of 53.71 °C and H_2O_2/C=C molar ratio of 1.72:1) (Borugadda and Goud, 2016). Meanwhile, the optimum conditions for epoxidized *Sapindus mukorossi* seed oil were found to be at an enzyme loading of 2.0 wt%, reaction time of 7 h, temperature of 50 °C, and H_2O_2/C=C molar ratio of 4:1 (Sun et al., 2011a).

In Malaysia, a development of environmentally friendly biolubricants originating from local sources has been focused on palm oil (Al-Zuhair et al., 2004; Talukder et al., 2010), palm kernel (Razak et al., 2015) and palm olein (Chew et al., 2008; Hassan et al., 2016; Jabal et al., 2014). However, less attention is being given to the utilization of palm stearin, which is formed during the production of palm cooking oil. Palm oil consists of about 50% saturated fatty acids (44% palmitic acid, 5% stearic acid and the balance of myristic acid) while palm stearin oil approximately 61.3% (55.11% palmitic acid, 4.32% stearic acid and the balance of myristic acid and lauric acid). Palm oil contains high concentration of unsaturated fatty acids of about 50% with 44% oleic acid, 13.4% linoleic acid and the rest is linolenic acid, in comparison with palm stearin oil of about 32% only (Montoya et al., 2014; Afifah et al., 2019).

In Malaysia, the production of palm stearin has been about 2.5 million tonnes per year since 2014 which showed that the stocks of this oil are always available as long as there is a demand for palm cooking oil in the market. In our previous work (Afifah et al., 2019), we found that palm stearin methyl ester which produce from the transesterification of crude palm stearin oil has a potential to be used as a green lubricating oil. However, palm stearin methyl ester exhibited poor wear properties compared with commercial lubricant oil. It was attributed to the poor oxidation stability originated from unsaturated fatty acid in the palm oil structure. In that context, the unsaturation in the palm stearin methyl ester is eliminated through epoxidation reaction using hydrogen peroxide and acetic acid and catalysed by immobilized lipase in the present study.

This study was aimed at investigating the effects of the lipase concentration, reaction time, reaction temperature, and molar ratios of acetic acid and hydrogen peroxide on the enzymatic epoxidation reaction. RSM and a Box-Behnken design were used as optimization tools to achieve the optimum reaction conditions, and to understand the relationship between the reaction variables. After that, tribological measurements were conducted to investigate the capability of the produced bio-lubricant to separate and protect two contacting metal surfaces under various sliding speed and temperature conditions.

2. Materials and methods

2.1. Materials

Palm stearin methyl ester (PSME) was generously given by KLK Bioenergy Sdn. Bhd, Malaysia. *Candida antartica Lipase B* with spherical bead size from 300 to 710 μm was procured from Purolite Corp. Analytical grade of 30% hydrogen peroxide (SYSTERM), glacial acetic acid (HmbG Chemicals), 5.0% sodium bicarbonate solution (Vchem), anhydrous magnesium sulphate (SYSTERM), potassium iodide (QRëC), cyclohexane (QRëC), Wijs solution (QRëC), 0.1 mol sodium thiosulphate solution (QRëC), hydrobromic acid (QRëC), crystal violet (QRëC) and potassium hydrogen phthalate (QRëC) were used as received.

2.2. Epoxidation reaction

creflux condenser, thermometer and magnetic stirrer. Prior to epoxidation reaction, *Candida anatartica* lipase was suspended into 30g of PSME and then the solution is stirred for 30 min. After that, glacial acetic acid is introduced into the solution and heated to the desired temperature. Hydrogen peroxide was dropped into the solution for every 30 min to avoid a chance of explosion due to exothermic reaction. Once the reaction completed, the solution was filtered first before neutralized with sodium bicarbonate solution and distilled water. Prior to analysis, the product which is epoxidized palm stearin methyl ester (EPSME) was dissolved in anhydrous magnesium sulphate to remove water and then filtered.

2.3. Analytical techniques

The relative oxirane conversion (ROC) was determined using the following expressions:

$$Yield\ of\ ROC\ (\%) = \frac{OOC_{exp}}{OOC_{theo}} \times 100 \quad (1)$$

where OOC_{exp} and OOC_{theo} are experimental and theoretical contents of oxirane oxygen. OOC_{exp} is identified according to AOCS official method Cd-9-57 using hydrobromic acid solution. Approximated 0.45 g of oil sample was mixed into acetic acid. Then around 0.1 ml of crystal violet was dropped into the solution and titrated directly with standardized hydrogen bromide solution. Then, the experimental amount of oxirane oxygen content was calculated using equation (2) as follows:

$$OOC_{\exp} = \frac{(V_b - V_s)\ x\ N\ x\ 1.60}{w} \quad (2)$$

where V_b and V_s represent the volume of standardized HBr solution required for titration of blank and epoxidized oil (ml) respectively, while w and N are the mass of oil sample (g) and normality of standardized HBr solution. OOC_{theo} of the epoxidized oil can be mathematically determined as follows:

$$OOC_{theo} = A_o \times \frac{IV_o/2A_i}{100 + \left[\left(IV_o/2A_i\right)A_o\right]} \times 100 \quad (3)$$

where IV_o is iodine value of vegetable oil before undergo epoxidation reaction, A_i and A_o represent atomic mass of iodine and oxygen, respectively. The iodine value was ascertained in accordance to AOCS official method Tg-1-64 by using Wijs technique. The epoxidized oil was dissolved into cyclohexane-glacial acetic acid mixture (20 ml) and Wijs solution (25 ml), and then stored the solution in a dark storage for 30 min. Later potassium iodide solution was added into the solution for about 20 ml followed with distilled water (100). Afterward, the solution is titrated with sodium thiosulphate solution 0.1 N. Afterward, iodine value of oil is calculated using the following equation:

$$IV = \frac{(V_b - V_s)\, x\, 0.1\, x\, 12.69}{w} \quad (4)$$

where V_b and V_s are volume of sodium thiosulphate solution (mL) required for titration of blank and epoxidized oil respectively, while w is the oil weight (g).

2.4. Experimental design and statistical analysis

The rate of epoxidation is governed by the molar ratios of carboxylic acid/C=C and H_2O_2/C=C, reaction time, reaction temperature and catalyst concentration. In the present work, these variables were preliminarily screened to identify the significant variables or variables using one half of a 2^{5-1} factorial design (Table S1 of supplementary data). The following ranges of experimental variables were chosen: temperature (30 °C–60 °C), reaction time (4 h–7 h), and enzyme concentration (2 wt% to 6 wt%), molar ratio of H_2O_2 (1.5 mol–4 mol), and molar ratio of acetic acid (0.2 mol–0.6 mol). The preliminary results indicated that the enzyme concentration, molar ratio of acetic acid/C=C and reaction temperature were the most significant variables that strongly influenced the ROC percentage or the epoxidized palm stearin methyl ester yield (EPSME) since the variables gave higher percentage of contribution of about 31.1%, 12.4% and 5.42%, respectively. While, the others variables, like reaction and molar ratio of hydrogen peroxide/C=C only contributed 3.0% and 0.2%. A maximum ROC of 63.93% was obtained under the following conditions: temperature of 59.6 °C, 5.52 wt% of enzyme, 1.51 mol of H_2O_2, and 0.2 mol of acetic acid. These conditions were then used as a reference for a one-factor-at-a-time study of the significant variables (Fig. S1) to determine new ranges that were narrower than previously for the optimization studies via response surface methodology. Then, a three-level-three-factor Box-Behnken statistical design was utilised to further maximize the ROC percentage. For the RSM for the epoxidation of PSME, the experimental ranges and levels of the significant variables were chosen, as listed in Table 1. The experimental design matrix, regression analysis and final optimization of variables were generated by Design Expert 11.1.0 software. A total of 14 designed experiments were conducted separately based on the design matrix with two centre points, as provided in Table 2. After that, a quadratic polynomial model was fitted to the experimental data acquired from each design to determine the optimum conditions for the epoxidation of PSME by using equation (5) as follows:

$$Y = \beta_0 + \sum_{i=1}^{n} \beta_i X_i + \sum_{i=1}^{n} \beta_{ii} X_i^2 + \sum_{i=1}^{n} \sum_{j<1}^{n} \beta_{ij} X_i X_j \quad (5)$$

where Y represent the yield of ROC %, β_0 β_i, β_{ii} and β_{ij} signify a constant, linear, quadratic and interaction regression coefficient terms respectively, while X_i and X_j are the independent variables.

Table 1
Experimental range and level of significant variables for epoxidation of PSME.

Variables	Symbol	Unit	Coded factor		
			−1	0	+1
Reaction temperature	x_1	°C	50	54	58
Acetic acid/C=C molar ratio	x_2	mol	0.15	0.18	0.21
Catalyst concentration	x_3	wt.%	5.98	6.99	8.0

2.5. Tribological measurement of biolubricants

Tribological characteristics of the produced lubricant were investigated using a fourball tribometer in accordance to ASTM D4172 (392 N, 1200 rpm, 75 °C). The temperature and sliding speed of the tribo-system were varied to from 75 to 95 °C and 1200–2000 rpm, respectively. The steel balls used for the experiments were made of chromium steel alloy (AISI) and having a diameter of 12 mm. Tribological performance of biolubricants were evaluated in term of friction and wear properties, where coefficient of friction was directly calculated by the tribometer software while wear scar diameter of the damaged surfaces were examined under optical microscope.

3. Results and discussions

3.1. Statistical analysis of epoxidation reaction of PSME

In the present work, Box-Behnken statistical method was employed for epoxidation of PSME. The significant variables were investigated, namely reaction time, C=C/acetic acid molar ratio and concentration of enzyme in order to understand their effect on the conversion of carbon-carbon double bonds into epoxy cyclic rings. Both experimental and predicted results were tabulated in Table 2 while Table 3 presents the results of ANOVA tests. The second-order polynomial equation obtained from experimental data to model the dependency of the ROC (%) response on the independent variables in terms of coded levels was presented as below:

$$Y = 92.52 - 13.95\, x_1 - 11.61\, x_2 - 10.32\, x_3 + 5.13 \times 1 \times_2 - 9.0 \times 1 \times_3 - 8.84 \times 2 \times_3 - 19.71\, x_1^2 - 15.81\, x_2^2 + 6.77\, x_3^2 \quad (6)$$

From the ANOVA results, it was found that the F-value for the polynomial model was 23.38, which indicated that the fitting of the model was statistically significant. There was only a 0.41% chance that the F-value could occur due to noise. Meanwhile, the p-value was used as a tool to determine the significance of each coefficient in the developed model. A p-value of less than 0.05 indicates that a model term is significant. According to the ANOVA results, the linear coefficients of x_1, x_2 and x_3, interaction coefficients of x_1x_3 and x_2x_3, and quadratic coefficients of x_1^2 and x_2^2 were significant model terms since their p-values were below 0.05. Besides that, the F-value of 1.38 demonstrated that the lack-of-fit was not significant relative to the pure error, thereby indicating that the model had a high precision and good reproducibility of experimental data. There was a 54.28% chance that a large F-value could occur due to noise.

The degree of precision of the model was further determined by evaluating the regression coefficient (R^2), predicted regression coefficient (pred-R^2), and adjusted regression coefficient (adju-R^2). A satisfactory statistical model should have an R^2 value that is approximately 1.0. In the present work, the large R^2 value of 0.9813

Table 2
The experimental design of significant variables and response value of ROC (%).

Run	A: Reaction temperature	B: Acetic acid/C=C molar ratio	C: Catalyst concentration	Response	
				ROC % (Experiment)	ROC % (Predicted)
1	54	0.15	7.5	92.46 ± 0.6	93.61
2	50	0.18	5.98	92.79 ± 0.1	94.85
3	54	0.21	7.5	55.52 ± 0.5	52.71
4	58	0.18	5.98	88.92 ± 0.1	84.95
5	50	0.21	6.74	55.60 ± 0.1	54.21
6	58	0.21	6.74	31.49 ± 0.6	36.57
7	54	0.18	6.74	88.73 ± 0.4	92.52
8	50	0.15	6.74	92.50 ± 0.2	87.69
9	58	0.15	6.74	48.35 ± 0.4	49.53
10	58	0.18	7.5	48.88 ± 0.5	46.31
11	54	0.21	5.98	91.10 ± 0.8	91.03
12	54	0.15	5.98	94.51 ± 0.5	96.57
13	54	0.18	6.74	96.39 ± 0.5	92.52
14	50	0.18	7.5	88.61 ± 0.4	92.21

Table 3
Regression analysis of variance (ANOVA) for % conversion of EPSME.

Source	Sum of Squares	Degree of freedom	Mean Square	F-value	p-value	
Model	6492.76	9	721.42	23.38	0.0041	Significant
x_1: Temperature	1557.38	1	1557.38	50.47	0.0021	
x_2: AA	1077.87	1	1077.87	34.93	0.0041	
x_3:Catalyst	851.61	1	851.61	27.6	0.0063	
x_1x_2	105.37	1	105.37	3.41	0.1383	
x_1x_3	323.82	1	323.82	10.49	0.0317	
x_2x_3	312.41	1	312.41	10.12	0.0335	
x_1^2	1242.68	1	1242.68	40.27	0.0032	
x_2^2	799.99	1	799.99	25.93	0.007	
x_3^2	146.61	1	146.61	4.75	0.0948	
Residual	123.42	4	30.86			
Lack of Fit	99.41	3	33.14	1.38	0.5428	Not significant
Pure Error	24.01	1	24.01			
Cor Total	6616.19	13				
R^2	0.9813					
Adjusted R^2	0.9394					
Predicted R^2	0.7451					
Adequate precision	12.7785					
Std. Dev.	5.55					
Mean	76.09					
C.V. %	7.3					

suggested that the model possessed a higher accuracy and could be used for the prediction of the experimental response value. Meanwhile, the pred-R^2 of 0.7451 was in reasonable agreement with the adju-R^2 of 0.9394 because the difference was less than 0.2. In addition, the adequate precision value, which was larger than 4.0, demonstrated that the signal-to-noise ratio was considered to be satisfactory. The coefficient of variance determines the ratio of the standard error of the estimate to the mean value of the response. The coefficient of variance of the model was 7.3%, which was lower than 10%, thereby confirming the high precision and reliability of the generated model. The coefficient of variance determines the ratio of the standard error of the estimate to the mean value of the response (Mashhadi et al., 2018).

3.2. Effect of independent variables on the percentage of ROC

In the present work, three different variables, namely the reaction temperature, acetic acid/C=C molar ratio, and catalyst concentration, were varied between 50 and 58 °C, 0.15–0.21 mol, and 5.98 to 7.5 wt%, respectively, as presented in Fig. 1 and Fig. S2 (Supplementary data). Fig. 1(a) demonstrates the influence of the reaction temperature, AA ratio (acetic acid/C=C molar ratio) and their mutual interaction on the yield of EPSME. From the 2D and 3D graphs, it was very conspicuous that the amount of acetic acid had a significant effect on the formation of oxirane rings because a slight increase in the molar ratio of acetic acid from 0.18 mol to 0.21 mol led to a considerable reduction of ~30% in the ROC percentage. In an epoxidation reaction, the presence of acetic acid is necessary to function as an active oxygen carrier, where the acetic acid will react with hydrogen peroxide to produce peracetic acid and water as side products. However, an excessive amount of acetic acid may trigger the rapid formation of water, which will result in undesirable reactions, such as i) the opening of the oxirane rings; and ii) hydrolysis (Kulkarni et al., 2013) which will eventually reduce the yield of epoxidized fatty acid, as was observed in the present work when the molar ratio of acetic acid was larger than 0.19 mol. A similar behaviour was observed for the reaction temperature, where the ROC of EPSME declined remarkably from 90% to 50% when the temperature was raised slightly from ~54 to 58 °C. Such an observation was mainly attributed to the adverse effect of the reaction temperature on the catalytic performance of lipase. Above 55 °C, the immobilized structure of the enzymes started to denature, thereby reducing their catalytic activity and eventually, the epoxidation rate. Therefore, the maximum conversion of EPSME was obtained at a lower acetic acid molar ratio range of ~0.155–0.18 mol and reaction temperature range of 50–54 °C.

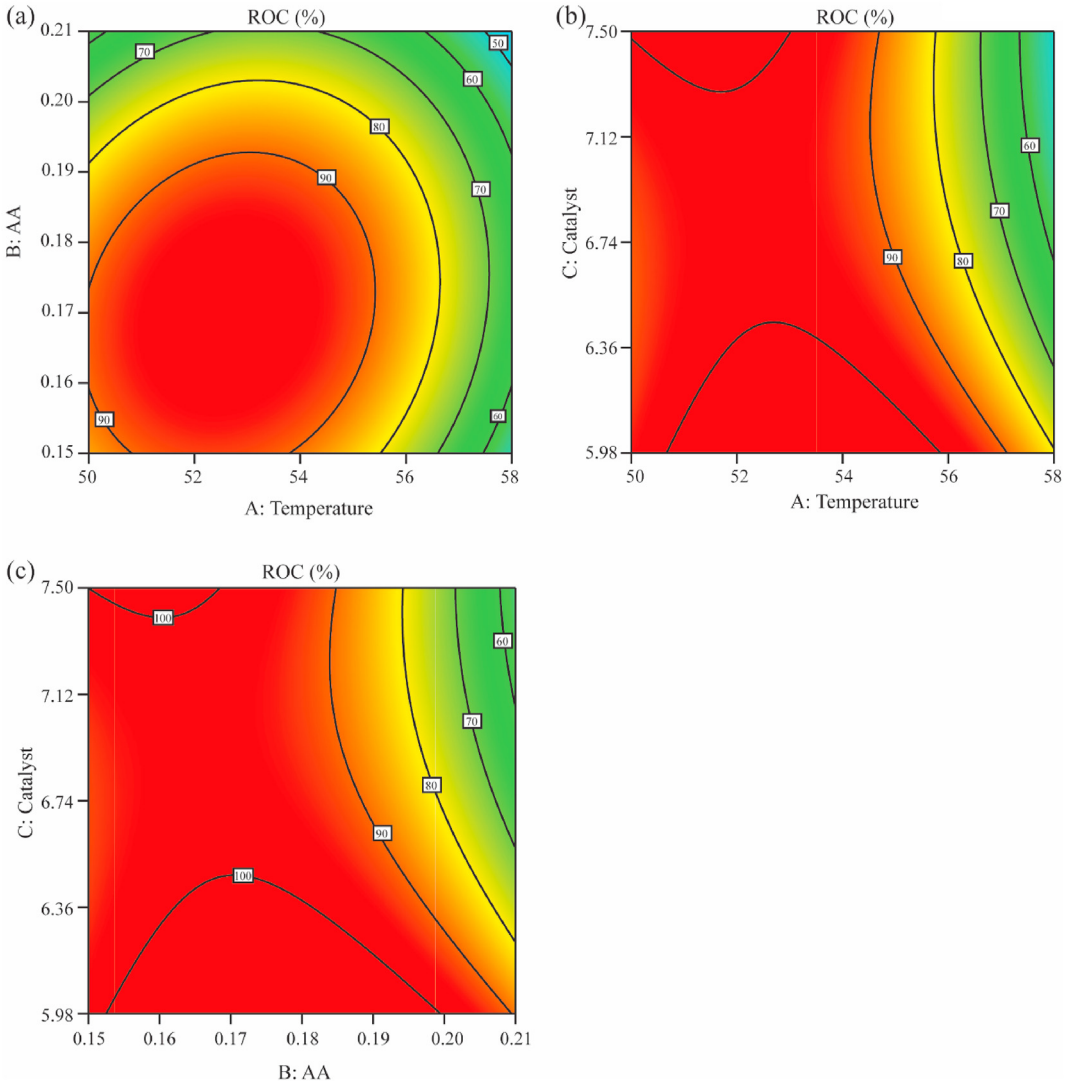


Fig. 1. Contour plots shows dependency of ROC yield on the (a) reaction temperature and acetic acid molar ratio (b) temperature and catalyst concentration (c) catalyst concentration and acetic acid molar ratio.

Fig. 1(b) shows the influence of the reaction temperature, catalyst concentration, and their mutual interaction on the epoxidation reaction of PSME. From the response surface plots, it was found that the maximum percentage of ROC was achieved at catalyst concentrations ranging between 5.98 and 6.36 wt%, and the ROC percentage appeared to have been slightly reduced when the enzyme concentration was increased beyond the range. This can be explained by the fact that the enzyme concentration is linearly related to the rate of formation of peracetic acid and epoxy rings. However, when the enzyme concentration is excessive in a reaction medium, there is the possibility that a simultaneous reduction may occur in the mass transfer and acceleration in the hydrolysis reaction, leading to retardation in the conversion rate (Dinda et al., 2016). A similar finding was also observed in other epoxidation reactions with vegetable oils such as karanja oil (Bajwa et al., 2016), corn oil (Sun et al., 2011b), soybean oil (Chen et al., 2017) and *Sapindus mukorossi* methyl ester (Sun et al., 2011a). Using the reaction temperature as another variable, it was noticed that the trend was similar with Fig. 2, where the ROC percentage declined with temperatures above 55 °C, although the enzyme concentration was increased at the same time. It appeared that a

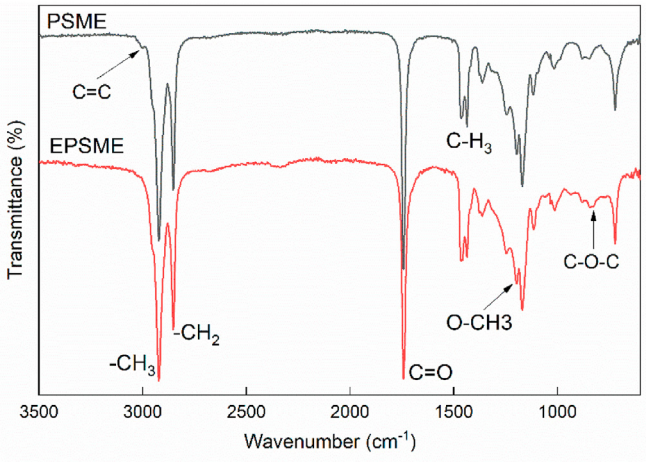


Fig. 2. FTIR spectra of PSME and its epoxide, EPSME.

high enzyme concentration was unable to enhance the percentage of conversion to epoxide rings even though the enzymes provided more active sites for the reaction. The interaction of the variables in the figure below suggests that the desired conversion of C=C bonds into oxygen oxirane rings was achieved at lower enzyme concentrations of 5.98–6.36 wt% and reaction temperatures of 51–55 °C.

The effects of the catalyst concentration, acetic acid/C=C molar ratio, and their interaction on the ROC percentage of EPSME are presented in Fig. 1(c). At lower enzyme concentrations, raising the molar ratio of acetic acid/C=C from 0.15 to 0.20 mol resulted in a slight enhancement of the ROC percentage of EPSME, while increasing the enzyme concentration at lower acetic acid/C=C molar ratios led to a decrease in the ROC percentage. Such behaviour was in mutual agreement with the results depicted in Figs. 2 and 3, thereby confirming the limited influence of both variables on the EPSME yield. Hence, it can be inferred that high concentrations of both the enzyme and acetic acid will have an adverse effect on the conversion of the double bonds into oxygen oxirane rings.

3.3. Optimization of significant variables

In the present study, the optimization was conducted based on the desired functions to determine the optimal epoxidation variables for PSME. According to the RSM optimization tools, the optimal s required to maximize the ROC percentage were a reaction temperature of 52 °C, catalyst concentration of 6.416 wt%, and 0.163 mol of acetic acid. Under these conditions, the ROC of PSME was predicted by the model to be 99.3%. To validate the accuracy of the predicted model, a confirmatory experiment was conducted in triplicate under identical conditions, and a maximum ROC of 98.9 ± 0.6% was achieved. The good agreement between the predicted and experimental values confirmed the accuracy and reliability of the developed model.

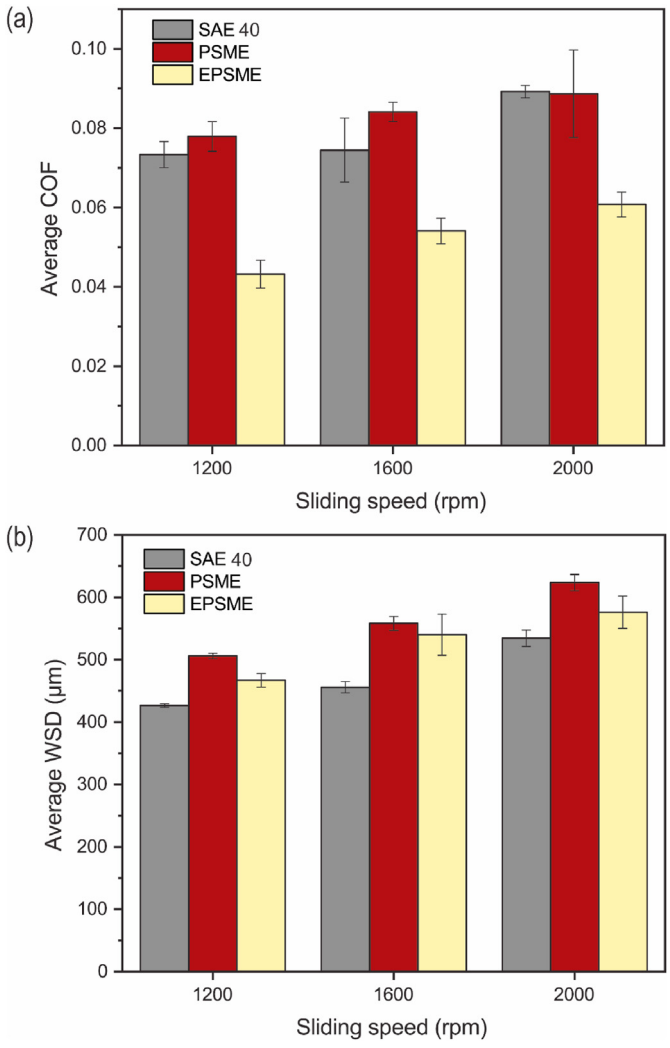


Fig. 3. Average (a) COF and (b) WSD of SAE 40, PSME and EPSME at different sliding speed conditions.

3.4. Characterization of final product

FTIR spectroscopy was used to monitor changes in the chemical structure of PSME before and after the epoxidation reaction, as depicted in Fig. 2, which consisted of two spectra: that of the raw material (PSME), and the modified PSME (EPSME). For the PSME, the presence of a peak at 3005 cm^{-1} corresponded to the stretching vibrations of the C=C double bonds. The disappearance of this peak in the EPSME spectrum showed that almost all the C=C double bonds had been transformed into oxygen oxirane rings due to the formation of a new peak at 823 cm^{-1}. The C-O-C stretching usually appears at the 820-850 cm^{-1} signal. Furthermore, from the EPSME spectrum, it was very conspicuous that the signals corresponding to the vibrations of methyl ester at 1195 cm^{-1} and 1435 cm^{-1} remained unchanged. Therefore, there was no evidence that side reactions of hydrolysis or the opening of rings took place during the epoxidation proce ss. The product of hydrolysis can be identified through the formation of a new peak at 1650 cm^{-1}, corresponding to carbonyl from the carboxylic acid group (López et al., 2009). while for the side reaction of ring-opening, a strong -OH stretching vibration of the hydroxyl group will appear around 3400-3500 cm^{-1} (Hazmi et al., 2013; Huang et al., 2015; Pagacz et al., 2019).

3.5. Tribology of biolubricants

In this study, the effects of sliding speed and temperature on the tribological performance of EPSME were investigated using fourball tribotester. During tribological test, the upper ball is held in a chuck rotating at a given speed, in this study from 1200 to 2000 rpm, while three lower balls are held stationary in a pot filled with the tested lubricant oil. The fixed applied or axial load is exerted on the upper ball and is transmitted to each of lower balls equally. During rotating process, a frictional force is measured by transducer that respond to the frictional force acting on the lower balls. Once finished, coefficient of friction or COF value is calculated using the following equation according to IP-239:

$$COF = \frac{Frictional\ torque\ (kg.mm)\ x\ \sqrt{6}}{3\ x\ Applied\ load\ (kg) x\ distance\ of\ balls\ (mm)} \quad (7)$$

Fig. 3 depicts the COF mean values between steel-steel tribo-contacts in the presence of PSME, EPSME and SAE 40 lubricants. SAE 40 is a mineral-based monograde engine oil that widely used in gasoline automotive engines of passenger cars, commercial vehicles and farm equipment. In general, the COF of all lubricants was seen to increase with increased sliding speed from 1200 rpm to 2000 rpm. It was found that EPSME exhibited the lowest COF values in all sliding speed conditions. For wear behaviour, the mean WSD of EPSME was slightly lower than its raw material; which was attributed to better oxidation stability. The presence of oxygen oxirane ring, in the backbone of PSME, offered better friction and wear characteristics.

Initially, both PSME and EPSME lubricants adhered to the metal

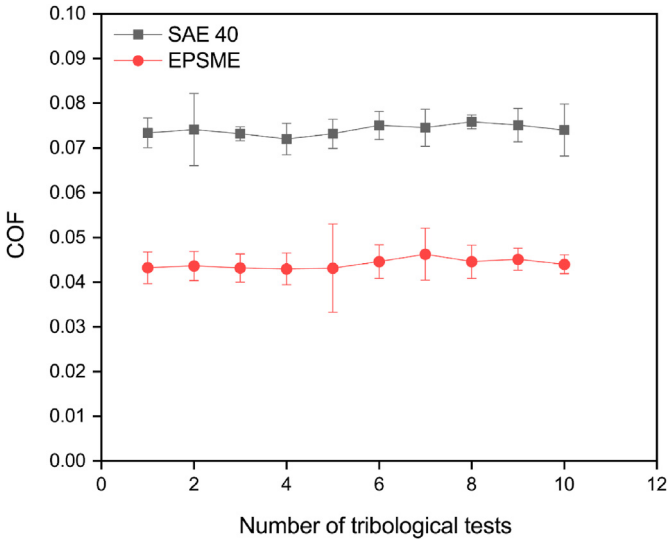


Fig. 4. COF of EPSME and SAE 40 as a function of number of tests.

surfaces through ester functional groups, which generated a formation of thick lubrication film that improved metal-to-metal separation. Increasing the sliding speed of the tribo-system could lead to a rise in interface temperature between rubbing metal surfaces. When the energy between the interfaces is high enough, carbon-carbon double bonds, which are located in the middle of the hydrocarbon, will be triggered to break up and unintentionally reduce the number of carbons within the chains. At this moment, a thick lubrication film will become thinner, thus reducing the strength of the lubrication film and its ability to separate metal surfaces. This phenomenon causes high COF values as shown in the PSME results. However, it is unlikely that the broken oxygen oxirane ring in EPSME would not affect the length of the hydrocarbon chain and the properties of the lubrication film. Therefore, modifying the unsaturated fatty acids in the vegetable oil could preserve a thick lubrication film for quite some time before it is degraded. A maximum enhancement of 44.5% was observed at 1200 rpm for a friction of 7.6% and wear properties of 2400 rpm. In addition, a stability test was conducted according to ASTM D4172 (392 N, 1200 rpm, 75 °C) in order to identify the friction behaviour of lubricants after being used repetitively. As shown in Fig. 4, it can be seen that COF values for both SAE 40 and EPSME remained nearly same up to ten tribological tests, around 0.0439 and 0.0741 respectively. The duration of each test was 1 h. From the obtained results, it can be inferred that the structure of EPSME not easily broken up to 10 times of used, thus could provide similar behaviour of lubrication film on the metal surfaces like a fresh EPSME.

Fig. 5 shows morphologies of the worn surfaces of steel balls lubricated with SAE 40, PSME and EPSME under various sliding speed conditions. The figure shows that steel surfaces lubricated with SAE 40 and PSME have many deep grooves, scratches and several material detachments. These features were formed when asperities of the counter surface penetrated into the current surface and caused micro-cutting along the sliding direction (Ma and Lu, 2010). Meanwhile, the formation of crevices and holes on the steel surfaces resulted from the adhesion process (Okonkwo et al., 2016). These observations suggest that the wear mechanism of

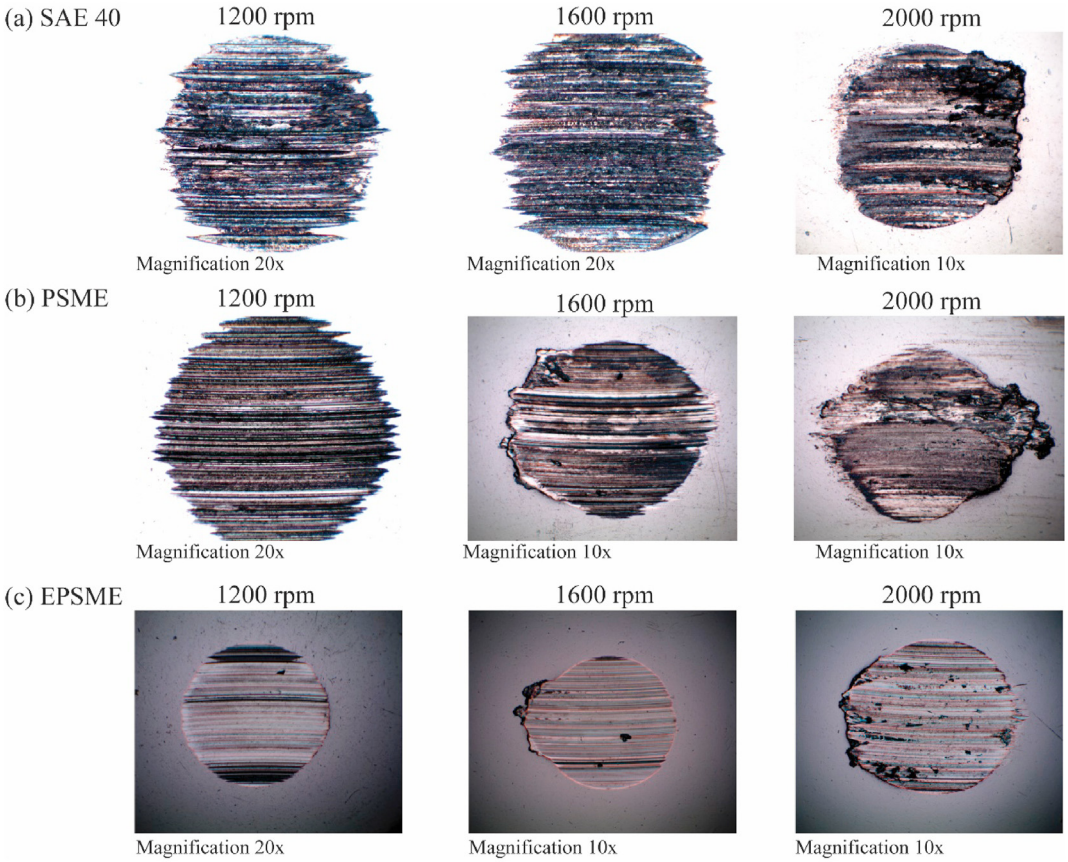


Fig. 5. Surface morphologies of worn steel lubricated with (a) SAE 40 (b) PSME and (c) EPSME after tested at different sliding speeds.

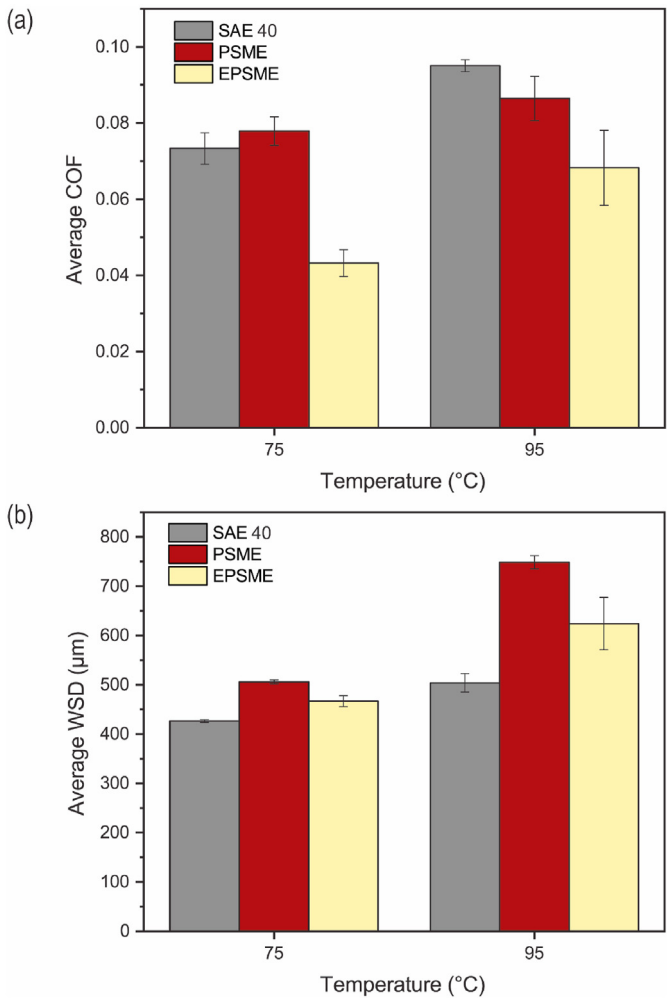


Fig. 6. Average (a) COF and (b) WSD of all lubricants at 75 and 95 °C.

these surfaces was dominated by both abrasion and adhesion wear. For EPSME, the presence of deep grooves was only profound at the top and bottom of the scar, thus showing a lower occurrence of abrasion wear.

As the sliding speed is increased to 1600 rpm, the occurrence of abrasion and adhesion wear became more pronounced for all lubricants because the steel surfaces were being gradually worn, thus forming a larger contact area (or scar). Consequently, more surface materials were being removed from the contact areas as debris. This indicates a very close asperity-asperity interaction that led to the rapid collisions. At high speed of 2000 rpm, it can be seen that there was a transition in the morphologies of the scar surfaces, which implied a transition from mild to severe wear mechanism; except for EPSME. Furthermore, formations of material build-up at the edge of the scar were significantly observed for SAE 40 and EPSME. It is believed that this behaviour was attributed to the rapid rise in localized frictional heating that tends to soften the surface and eventually cause the flow and sticking of material in particular areas (Gupta and Harsha, 2018; Pearson et al., 2013). However, SAE 40 gave the lowest mean WSD value; its surface damage was apparently worsened compared to EPSME. This showed that epoxidized vegetable oil is highly capable of reducing collisions between asperities whilst maintaining separation of metal-to-metal contact.

Means of COF and WSD, as a function of temperature for all lubricants, are plotted in Fig. 6. It was found that the COF of all lubricants increased with temperature and a similar trend was also shown for wear behaviour. From the results, EPSME always demonstrated lower COF and WSD, compared to PSME. It is well known that augmenting the temperature of a tribo-system spontaneously increases the rate of oxidation of lubricating oil. The low COF and WSD numbers of SPSME, compared to PSME, indicates efficient formation of a thicker protective film on steel surfaces. This is shown in Fig. 7, where the scar morphologies of steel surfaces, after testing at 95 °C. The damaged surface consists of fewer deep grooves and material transfer than PSME. From the remarkably good lubricating and wear properties produced by EPSME, it can be inferred that epoxidized oil has the potential to replace the role of mineral oil as base oil in lubricant formulation.

4. Conclusion

An environmentally friendly palm stearin oil-based lubricant was successfully synthesised through epoxidation reaction. The optimum epoxidation variables that gave rise to a maximum relative conversion of 98.9 ± 0.6% of PSME were as follows: reaction temperature of 52 °C, catalyst concentration of 6.416 wt%, and 0.163 mol of acetic acid. The tribological results show that modifying the chemical structure of vegetable oil has a prominent effect on the tribological characteristics of the oil. A reduction in friction and wear properties was observed in each tribological experiment condition. The utilization of an environmentally benign local source by transforming it into a high value-added product will not only be economically attractive but more importantly could help to reduce the effects of global warming while improving the efficiency of engineering systems. The greater the reduction achieved in both tribological properties, the greater the energy that could be saved worldwide.

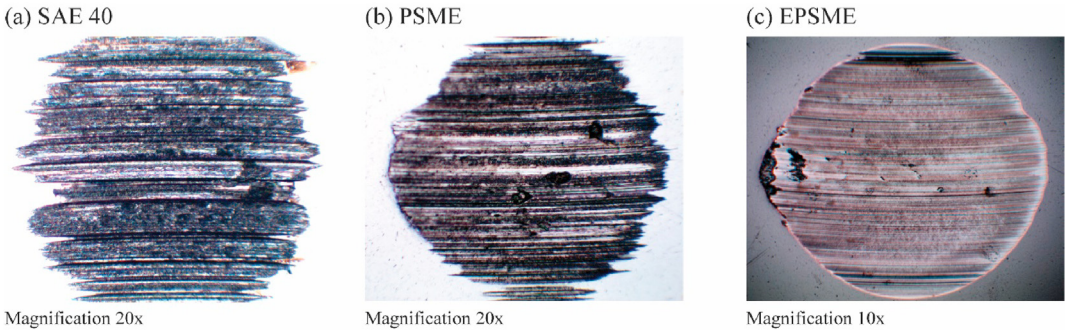


Fig. 7. Optical micrographs of (a) SAE 40 (b) PSME and (c) EPSME at 95 °C.

CRediT authorship contribution statement

A.N. Afifah: Conceptualization, Writing - original draft, preparation. **S. Syahrullail:** Methodology, Supervision. **Nur Izyan Wan Azlee:** Software, Validation. **A. Majid Rohah:** Methodology, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jclepro.2020.124320.

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